Codebase Revisit: Default Strategy Modularity and Rule Boundaries

Tactix Development Project

8 August 2026

Executive summary

The revisit preserved the shipped solver trajectory while making the strategy boundary substantially clearer. DefaultPolicy now contains the CDCL/search strategy and composes a dedicated DefaultInprocessScheduler for macro phase order, clocks, budgets, watermarks, delays, cursors, telemetry, and Factor/BVE coupling. Factor owns a neutral selection interface, and Factor and Substitute now report facts through kernel events instead of importing or invoking policy code. The high-volume source commentary was removed: comment-only lines in the three original hotspots plus the extracted scheduler fell from 1,542 to zero.

The refactor is trajectory-identical on 399 independently checked comparisons, passes the complete Debug and Release suites, and has no detected structural performance regression. The recommended next step is to resume the Kissat default-parity campaign one N4 row at a time through the new scheduler boundary, starting with SP-COMPACT-1 unless another row is explicitly selected.

1 Implementation and design decisions

1.1 Composed inprocessing scheduler

The new `src/policies/default_inprocess.hpp` owns the complete mutable macro-scheduling state and exposes a single test/preset surface through `DefaultPolicy::inprocess()`. DefaultPolicy retains branching, restart/reduce, mode switching, rephase/reorder, glue/decision-rate observations, and the search loop. The default-policy implementation shrank from 2,640 lines to 1,870 lines including the extracted header; its public header shrank from 586 to 148 lines.

1.2 Proof-rule versus strategy boundary

`FactorSelection` is defined by the Factor rule module. `FactorMatchPolicy` adapts that rule-owned interface into the policy system. The kernel holds only a scoped non-owning pointer to the neutral interface. Factor emits fresh-variable events and Substitute emits substituted and retirement events. Policies consume those events and apply heuristic reactions; the C API deliberately discards them as neutral advice. A new CTest source guard rejects policy imports and heuristic calls from every macro formula rule while retaining the four search-micro exceptions.

1.3 Comment cleanup

The cleanup deleted comment-only prose from `default_policy.cpp`, `default_policy.hpp`, and `kernel.hpp`; the extracted scheduler also contains no comment-only blocks. Executable tokens in

the mechanical pass were unchanged. Concise inline annotations and the calculus, implementation notes, design documents, tests, and parity ledger remain the owners of invariants and provenance.

2 Evidence

Gate	Result	Provenance
Debug CTest	60/60 passed	local Debug build
Release CTest	60/60 passed	local Release build
Source boundary	clean	`rule_policy_boundary` CTest
Identity	399 equal, 3 symmetric timeouts	SLURM 1592226
Focused fuzz	five 300-case legs, 0 failures	SLURM 1592227
Regression corpus	14/14, 0 failures	SLURM 1592227
Tier B	both 25/28, clean	run `20260808-203252-B-revisit-before`

The identity gate compared exit code, status/model lines, every non-timing counter line, and proof bytes across generated cases and the regression corpus under default DRAT, default LRAT, and an environment arm. Each completed result was then checked independently as a SAT model or UNSAT proof. Binary hashes were `ff9da623...` before and `6ae4120d...` after.

The focused fuzz gate covered default DRAT (299 verdicts), default LRAT (299), random LRAT (254), targeted Factor LRAT (256 and 80,010 target applications), and targeted Substitute LRAT (253 and 56,668 target applications). Timeouts were reported as lost coverage, not passes.

The same-node Tier B window produced PAR-2 93.21 before and 93.14 after, with identical statuses and rule counters on all 28 instances and all 13 UNSAT proofs checked in each arm. The runner correctly classified the headline as undecided because one instance reverses its sign. Excluding that named unstable mover, the 24 paired solved instances have median after/before runtime ratio 1.0026 (+0.26 percent), inside the pre-registered +5 percent ceiling.

3 Limitations and incidents

Three early identity submissions (1592221, 1592222, 1592224) were intentionally cancelled because the newly extended validator mishandled empty SAT models and drat-trim's intentional nonzero exit for trivial UNSAT. Diagnostic job 1592225 retained one case, showed `s VERIFIED`, and proved the two proof files byte-identical. The corrected gate then passed. These were harness defects, not solver failures.

The Tier B array was submitted with an unnecessary throttle of eight. An attempt to remove it raced with completion; all tasks had already been admitted, with only two running and none pending. Future arrays should be unthrottled unless the runner or experimental design specifically requires a cap.

The implementation creates a semantic scheduler component, but its method definitions remain in `default_policy.cpp`. A future purely mechanical translation-unit split is possible; it should be undertaken only if navigation remains a problem, under the same identity gate. No scheduling default, rule identifier, C API, proof format, or output schema changed.

4 Next actions

1. Resume Kissat-default parity through DefaultInprocessScheduler, taking one remaining N4 row at a time; SP-COMPACT-1 is the recommended first row.

2. Preserve the new rule-policy source guard and event tests with every macro rule change.
3. Run the planned Tier D shakedown, full Tier D campaign, and T3 gate after the remaining parity rows close.